



MOHAMED SHERIFF MOHAMED WASEEM

BSc. In Electrical & Electronics Engineering |
Electronics, Industrial Automation, PLC/SCADA & IoT |
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PROFESSIONAL PROFILE

Electrical and Electronics Engineering graduate from SLIIT, with academic coursework completed and awaiting convocation in 2026. Hands-on industrial training experience in water treatment plant operations, instrumentation, SCADA monitoring, motor and pump systems, VFDs, relay operations, IoT automation, embedded systems, PCB design, and dashboard-based monitoring. Strong interest in electronics, PLC/SCADA, pneumatics, electro-pneumatics, industrial automation, and IoT-based control systems, with practical exposure to ESP32, Raspberry Pi, MQTT, sensors, actuators, relays, and control panels.

EDUCATION

BSc (Hons) in Electrical and Electronic Engineering | Sri Lanka Institute of Information Technology (SLIIT)

Malabe, Sri Lanka | Graduating 2026 | Status: Academic coursework completed; awaiting convocation.

G.C.E. Advanced Level (A/L) - Physical Sciences | Zahira College Kalmunai
Sri Lanka | Dec 2018

G.C.E. Ordinary Level (O/L) | Al-Jalal Vidyalaya
Sainthamaruthu | Dec 2015

EXPERIENCE

Industrial Trainee - Electrical, Instrumentation & SCADA | Zhongnan SZSG JH Joint Venture Company, Ambathale Water Treatment Plant Project

Ambathale, Sri Lanka | Jun 2024 - Aug 2024

- Gained practical exposure to pump house operations, SCADA-based monitoring, flow and pressure monitoring, remote monitoring, and manual inspection/log keeping.
- Participated in or observed insulation resistance testing for cables and motor windings, pump vibration checks, pump cooling water checks, and surge vessel level maintenance.
- Observed and documented instrumentation such as pressure transmitters, differential pressure transmitters, ultrasonic level transmitters, Prosonic sensors, Liquiline systems, and AUMA actuators.
- Developed understanding of motor troubleshooting, power supply issues, relay operations, control panels, VFD fundamentals, and HSE practices in a large-scale utility environment.

Industrial Trainee - IoT, Embedded Systems & Automation | Divor Automations (Pvt) Ltd

Pannipitiya, Sri Lanka | Jun 2025 - Sep 2025

- Worked with IoT device development, PCB design/fabrication, embedded programming, panel board assembly, industrial automation prototypes, and SCADA/HMI-related applications.
- Supported Hydroponic Grow Box activities including sensor calibration, PCB design, automatic lighting/irrigation control, and ESP32-based IoT integration.
- Assisted IoT dashboard projects using sensor-based data collection, MQTT communication, and real-time production data visualization concepts.
- Designed and tested PCB schematics/layouts using EasyEDA; practiced through-hole and SMD soldering, hot-plate soldering, continuity testing, and hardware debugging.

PROJECTS

Real-Time Automated Hydroponic Cultivation System using IoT and Sensors | Final Year Project

2026

GitHub

- Implemented a multi-node IoT automation system using ESP32 sensor/actuator nodes, Raspberry Pi 5, Mosquitto MQTT, Server.js backend, and a web/mobile monitoring interface.
- Configured Raspberry Pi 5 as the local server and central controller; supported MQTT broker setup, JSON payload processing, backend logic, dashboard integration, Docker deployment, VNC access, and Cloudflare/domain-based remote access.
- Integrated real-time monitoring for pH, TDS, water level, temperature, humidity, actuator states, and ESP32 online/offline status with separate dashboard cards for each node.
- Developed control logic concepts for water filling/draining, valve-pump coordination, pH-based nutrient dosing, manual mode, and emergency stop safety control.

Armyworm Pest Detection and Ultrasonic Repeller System | Design Project

GitHub

2025

- Designed an intelligent agricultural pest detection and repelling prototype using Raspberry Pi 5, Raspberry Pi Camera, servo motor scanning, Python, OpenCV, TensorFlow Lite/Teachable Machine, and an ultrasonic repeller circuit.
- Implemented image-based detection for armyworm identification and triggered a piezoelectric ultrasonic actuator through GPIO for non-chemical pest control.
- Constructed a Colpitts oscillator-based ultrasonic repeller circuit and validated detection-to-response operation in controlled simulation tests.

TECHNICAL SKILLS

Programming & Embedded: Embedded C / Arduino, Python, Java

Microcontrollers & Platforms: ESP32, Arduino boards, Raspberry Pi 5, STM32, NodeMCU, ESP8266

Automation & Control: SCADA/HMI exposure, PLC fundamentals, relay logic, VFD fundamentals, control panels, motor/pump systems

IoT & Networking: MQTT, Mosquitto broker, Wi-Fi, dashboards, remote monitoring, Cloudflare Tunnel, VNC, Docker basics

PCB & Electronics: EasyEDA, schematic design, PCB layout, Gerber generation, SMD/THT soldering, hot plate soldering, continuity checks

Testing Equipment: Digital multimeter, oscilloscope, power supply, insulation resistance tester, spectrum analyzer exposure

Sensors & Actuators: pH, TDS, ultrasonic level, DHT sensors, pressure/flow/level instrumentation, solenoid valves, relays, pumps

CORE TECHNICAL COMPETENCIES

Electronics & Embedded Systems: ESP32, Arduino, Raspberry Pi, sensors, relays, actuator control, signal conditioning, soldering, PCB testing

Industrial Automation: control panels, relay logic, motor starters, VFD fundamentals, HMI/SCADA exposure, automation troubleshooting

PLC/SCADA & Control: PLC fundamentals, SCADA monitoring, process parameters, alarms, real-time dashboards, manual/auto control concepts

IoT & Communication: MQTT, Wi-Fi, JSON data handling, IoT dashboards, real-time monitoring, cloud/remote access concepts

Pneumatics & Electro-Pneumatics: solenoid valves, relay-controlled actuators, pneumatic/electro-pneumatic circuit fundamentals

Electrical & Instrumentation: motors, pumps, insulation testing, transmitters, pressure/flow/level monitoring, panel inspection

ADDITIONAL INFORMATION

References: Available upon request